Large-scale traffic flow reconstruction from heterogeneous mobility sensors

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Abstract

Regional traffic reconstruction must combine observations that differ in spatial support, sampling and independence from the model. We reconstruct hourly link flows on the road network of Emilia-Romagna, Italy, by driving an agent-based simulation with mobile-phone origin–destination activity, informing route choice with GPS-derived speeds and road-class usage, and correcting the simulated counts locally with inductive loop detectors. The reconstruction is assessed with three independent tests: hourly profiles against GPS probes on all matched links of two campaigns, before and after the COVID-19 restrictions; daily volumes against loop detectors not used to calibrate route choice; and detectors withheld from the correction. Simulated and observed traffic peak at the same hours, and the reconstructed profiles have a median error of about one third of the mean hourly share, but they are less accurate than the injected demand profile and a mean detector profile. GPS-informed route choice, calibrated in one province, improves the rank agreement of link volumes with detectors in the rest of the region in both campaigns, while the absolute volume level remains about half of the observed one. The detector correction improves hourly profiles up to a few links away but does not transfer volumes to withheld detectors. The results identify which properties of a regional traffic field heterogeneous observations can support and which require further data.

Keywords: traffic reconstruction, mobile-phone demand, inductive loop detectors, GPS probes, route choice, agent-based simulation

1 Introduction

Mobile-phone records provide a geographically broad view of travel demand. Their regularities in space and time [1] have been used to estimate origin–destination (OD) matrices by time of day and trip purpose [2–4], and passively collected data are increasingly part of transport modelling practice [5–7]. Converting activity between areas into traffic on roads, however, requires assumptions about vehicle use, trip endpoints and routing, and phone-based samples carry their own ownership and activity biases [8]. Previous studies have combined mobile-phone demand with road networks and assignment models to estimate road usage and congestion [9–11], building on the equilibrium-assignment tradition [12]. Our group has studied GPS and mobile-device mobility in Italian cities, including Bologna and Rimini in Emilia-Romagna [13–18]. The open question for a regional application is what can be established independently about the resulting link counts and their variation through the day.

Fixed detectors and vehicle probes address different parts of this question. Detectors measure passages at a small set of road sections and have long been used to constrain OD estimation [19–21] and to calibrate the demand of agent-based simulations [22]. GPS observations cover additional roads but sample a fleet whose inclusion probability need not be constant between links or hours [23–26]. Traffic-state estimation and multi-sensor fusion therefore depend on both the physical model and the observation process [27–32]. Sensor location and flow observability place further limits on what sparse counts can identify [33–36]. In particular, reproducing the count of a detector already used for correction is not evidence of accuracy elsewhere.

These issues remain relevant alongside recent methods. Telecom–vehicle fusion has been studied for prediction away from cameras [37]; probabilistic estimation targets sensor-free traffic states [38]; and sparse-detector scaling methods estimate network traffic states from limited stationary coverage [39]. Uncertainty in demand inputs propagates through every stage of such models [40, 41], and simulation outputs require statistical validation against data not used for calibration [42, 43]. Simple historical profiles are also strong benchmarks for temporal traffic patterns [44, 45].

We reconstruct hourly flows on the regional road network of Emilia-Romagna from January 2020 mobile-phone OD activity, the `minimocas` agent-based simulator on an OpenStreetMap-derived graph, the regional inductive loop detectors (called coils below) and Viasat GPS probes. The probes play two separate roles: in one province they provide speeds and road-class usage for route choice; everywhere they provide independent hourly profiles. The contribution is a regional assessment that keeps three questions apart: whether the hourly shape of traffic is reproduced, whether the spatial distribution of volumes is reproduced, and how far local detector information propagates through the network. Each question is answered with data that did not enter the corresponding modelling step, and the reconstruction is compared with simple baselines throughout.

2 Results

2.1 Regional inputs

The regional network connects an east–west urban corridor, the Adriatic coast and Apennine roads (Fig. 1). The model graph has 3,925 directed polylines and 2,548 nodes; matching to OpenStreetMap assigns 709 km to motorways, 326 km to trunk roads, 2,629 km to primary, 4,673 km to secondary and 2,863 km to tertiary roads. Mobile-phone activity is aggregated to municipality-level OD relations and averaged separately for weekdays and weekends (Fig. 2). The hourly OD labels refer to local time and to the start of the hour: the weekday OD profile agrees best with the hourly profile of the MTS coils at zero lag (correlation 0.96, against 0.74 with a one-hour shift). Four endpoint classes represent observed OD pairs and residual demand with one or both endpoints sampled by the model.

2.2 Reconstructed field and local detector correction

The simulator routes agents over directed links and records, for each hour, the distinct agents present on each link. On 10 February 2020 the raw reconstruction on the link of coil 282 follows the daily shape of the counter

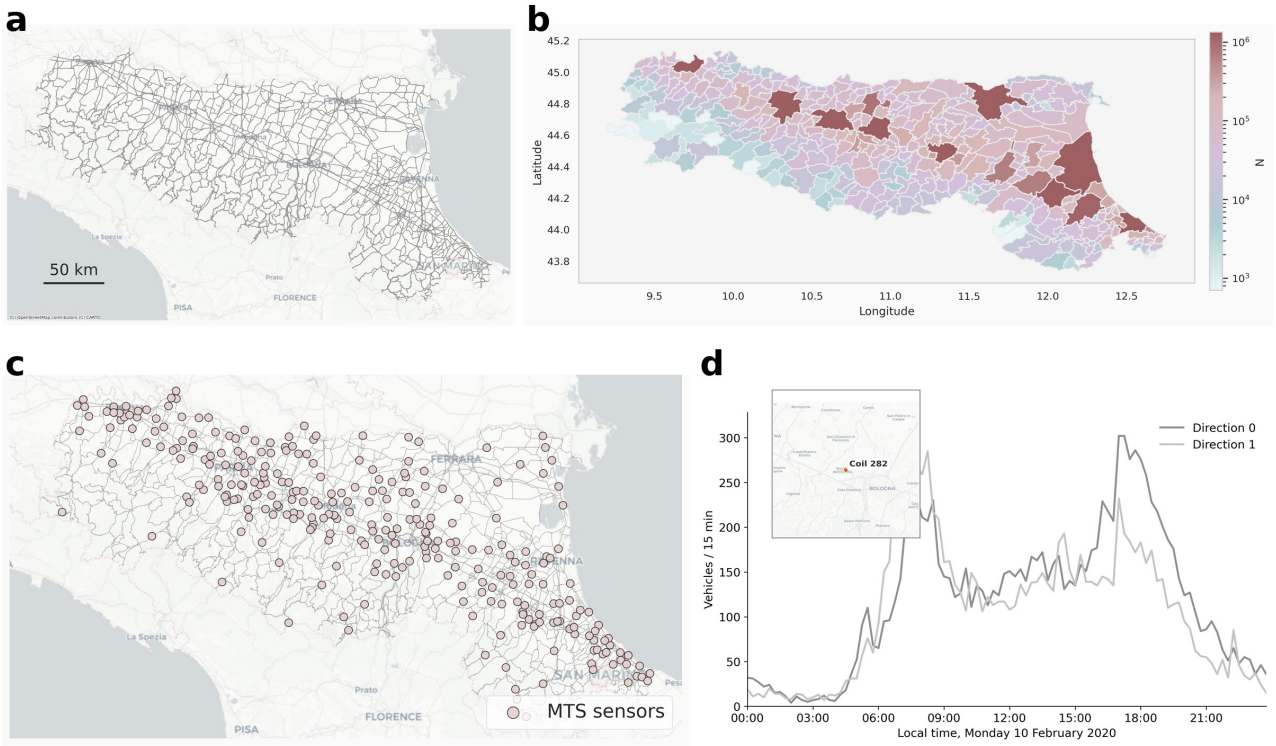


Fig. 1 Regional domain and observational inputs. (a) Regional road graph on a CartoDB Positron basemap; scale bar, 50 km. **(b)** Municipality-level mobile-phone OD activity in January 2020, logarithmic scale. **(c)** Locations of the MTS inductive loop detectors. **(d)** Directional counts of coil 282 on 10 February 2020 in local time, with its location in the inset.

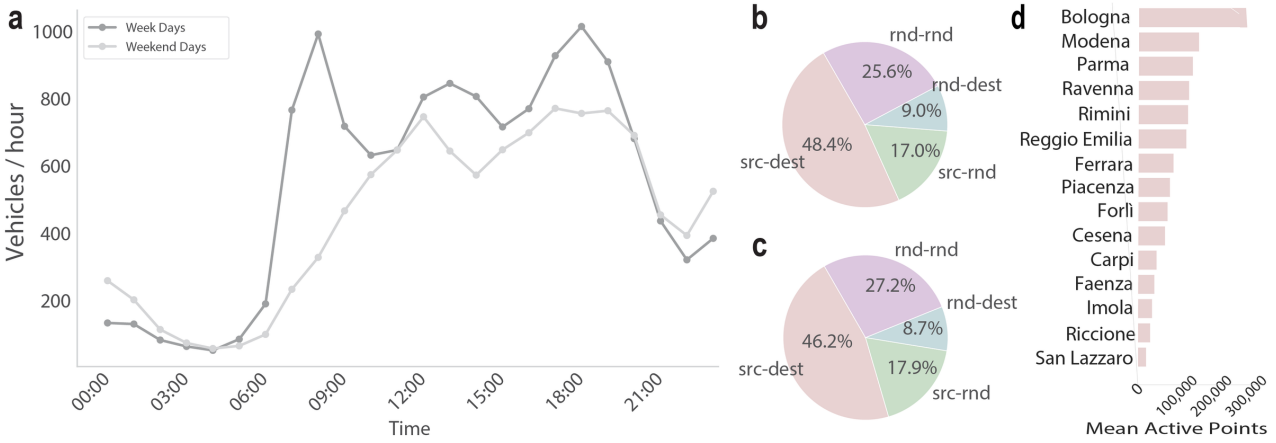


Fig. 2 Mobile-phone demand. (a) Mean hourly weekday and weekend activity in January 2020. **(b,c)** Weekday and weekend shares of source–destination, source–random, random–destination and random–random demand. **(d)** Activity of the 15 most active municipalities.

(hourly correlation 0.86) with 6,232 agents against 21,935 observed passages; after assimilation the link carries 21,808 vehicles (Fig. 3a). Agreement at an assimilated detector is a consequence of the correction and is not used as validation below. The daily map shows the reconstructed field over the whole network (Fig. 3b).

2.3 GPS-informed route choice

Two thirds of the injected demand belongs to the random-endpoint classes, whose agents choose the next link at each node with transition probabilities rather than following a best path; with uniform probabilities they spread over the network irrespective of road hierarchy. We used the raw GPS traces of private vehicles in the province of Forlì–Cesena (February 2020) to inform route choice with two road-class properties (Table 1): the off-peak median speed of the matched links, which sets link speeds and travel times, and the median daily

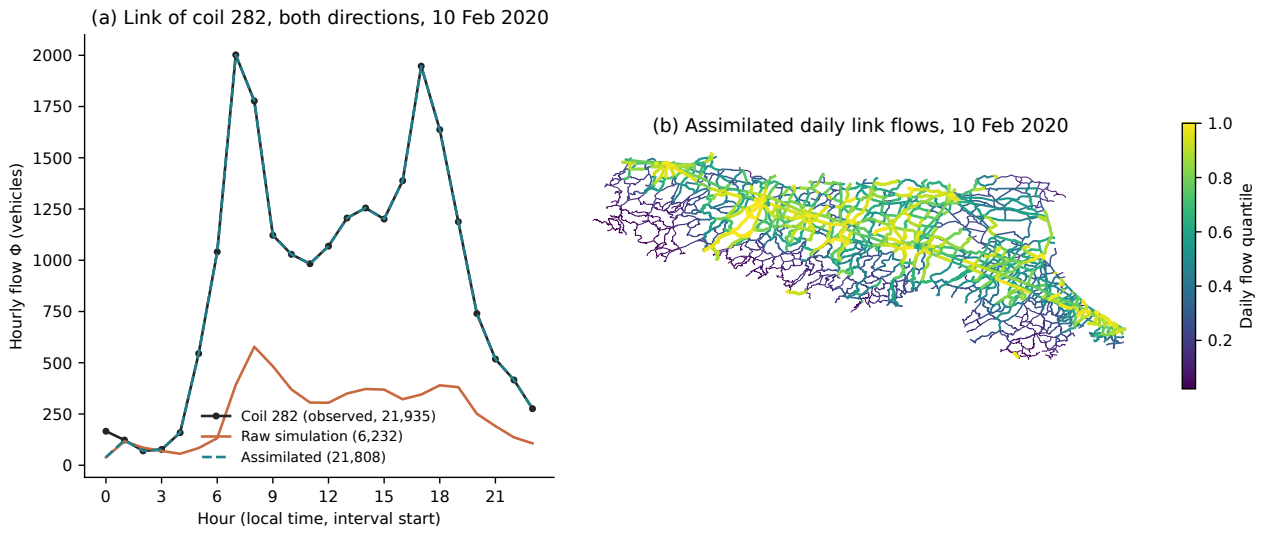


Fig. 3 Reconstructed field and local correction. (a) Hourly counts of coil 282 (both directions) and raw and assimilated reconstruction on its link, 10 February 2020, local time; totals in parentheses. **(b)** Assimilated daily link flows, coloured and scaled by flow quantile.

number of probes per covered link, which sets the transition weights of random-endpoint agents and a usage term in the best-path cost (Methods). Link speeds of the same roads in June 2021 have a rank correlation of 0.98 with those of February 2020.

Table 1 Road-class properties estimated from private-vehicle GPS traces in Forlì–Cesena, February 2020: median off-peak speed of links with at least 50 matched points (interquartile range in brackets) and median daily probes per covered link. Both are applied to all links of each class in the region.

Road class	Links (speed)	Speed (km/h)	Links (usage)	Probes per day
Motorway	44	123 [120–124]	61	130
Trunk	75	83 [76–95]	76	79
Primary	151	55 [46–64]	160	109
Secondary	245	51 [46–60]	248	36
Tertiary	193	53 [43–59]	227	21
Unclassified	42	49 [36–54]	57	14

Among the route-choice variants simulated for 10 February 2020 (Supplementary Section S3), we selected the one with the highest rank correlation between simulated link volumes and GPS link counts in Forlì–Cesena (0.54, against 0.47 for shortest-distance routing). It raises the share of vehicle-kilometres on motorways, trunk and primary roads from 0.39 to 0.48. All results below refer to this configuration; detectors outside Forlì–Cesena played no role in the selection.

2.4 Hourly profiles on all GPS-matched links

Viasat probes did not enter demand generation or detector correction. Matched links cover mainly the Forlì–Cesena area (Fig. 4). Of 930 matched links in 2020 and 927 in 2021, 821 and 818 have positive totals in both data sources. The campaigns contain eight days in 2020 (10–17 February, six weekdays and a weekend) and seven in 2021 (28 June–4 July, five weekdays and a weekend), each simulated with the OD matrix of its day type. The ratio between probes and simulated vehicles has a median of 1.4% in 2020 and 1.2% in 2021 and spans several orders of magnitude (Fig. 4c), so we compare hourly shares, which do not depend on it.

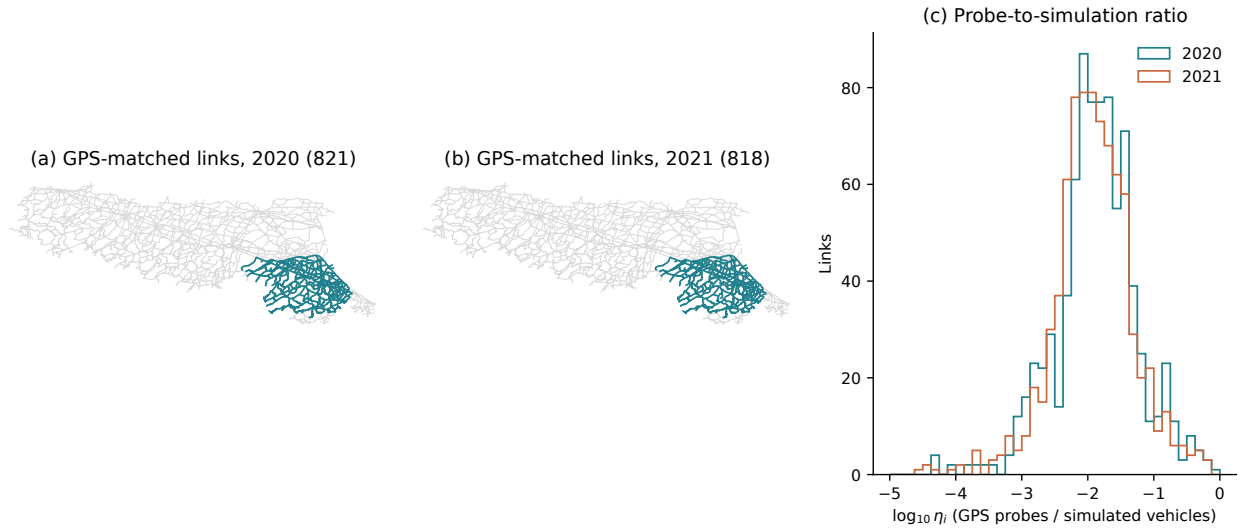


Fig. 4 GPS coverage. (a,b) Usable GPS-matched links (number in parentheses) over the regional graph, 10–17 February 2020 and 28 June–4 July 2021. **(c)** Distribution of the link ratio η_i between probe counts and simulated vehicles, Eq. (3).

Simulated and observed traffic peak at the same hours: pooled over all usable links, both have their maximum at 18:00 in both campaigns, and the pooled profiles differ by 0.66 and 0.64 percentage points (pp) per hour on average (Fig. 7a,b). At link level, the median error of the hourly shares is 1.41 pp in 2020 and 1.50 pp in 2021, about one third of the mean hourly share of 4.17 pp, with median hourly correlation 0.81 and 0.76 (Table 2, Fig. 5). The error decreases with the number of probes: on links with at least 200 probes per campaign, 511 in 2020 and 470 in 2021, the medians are 1.10 and 1.18 pp.

Table 2 Median hourly-share MAE (pp) on all usable GPS-matched links: raw and assimilated reconstruction, and three profiles applied to every link (injected OD demand, mean MTS detector profile of the same days, uniform). Weekends contain two days per campaign.

Campaign	Days	Links	Raw	Assimilated	OD	MTS	Uniform
2020	all (8)	821	1.52	1.41	1.27	1.17	2.71
	weekday (6)	817	1.68	1.60	1.41	1.33	2.87
	weekend (2)	813	2.17	2.11	2.01	1.90	2.97
2021	all (7)	818	1.50	1.50	1.30	1.21	2.57
	weekday (5)	816	1.62	1.60	1.44	1.40	2.71
	weekend (2)	803	2.36	2.33	2.19	2.03	2.94

Two fixed profiles set these values in context. The hourly profile of the injected OD demand has median errors of 1.27 and 1.30 pp, and the mean profile of the MTS coils on the same days 1.17 and 1.21 pp; the reconstruction is more accurate than the OD profile on 31% and 28% of the links and than the MTS profile on 24% and 23%. Network loading therefore does not sharpen the timing of the demand: travel times and the hourly accounting of agents that remain on a link for more than one interval broaden the simulated profiles. Separating weekdays and weekends raises all errors, because each subset has fewer probes, and the ordering of the methods is unchanged.

2.5 Link volumes at detectors

The raw reconstruction never uses the coils, so all coils with complete data provide an independent test of daily volumes. Outside Forlì–Cesena, the rank correlation between simulated and observed daily volumes is 0.36 in

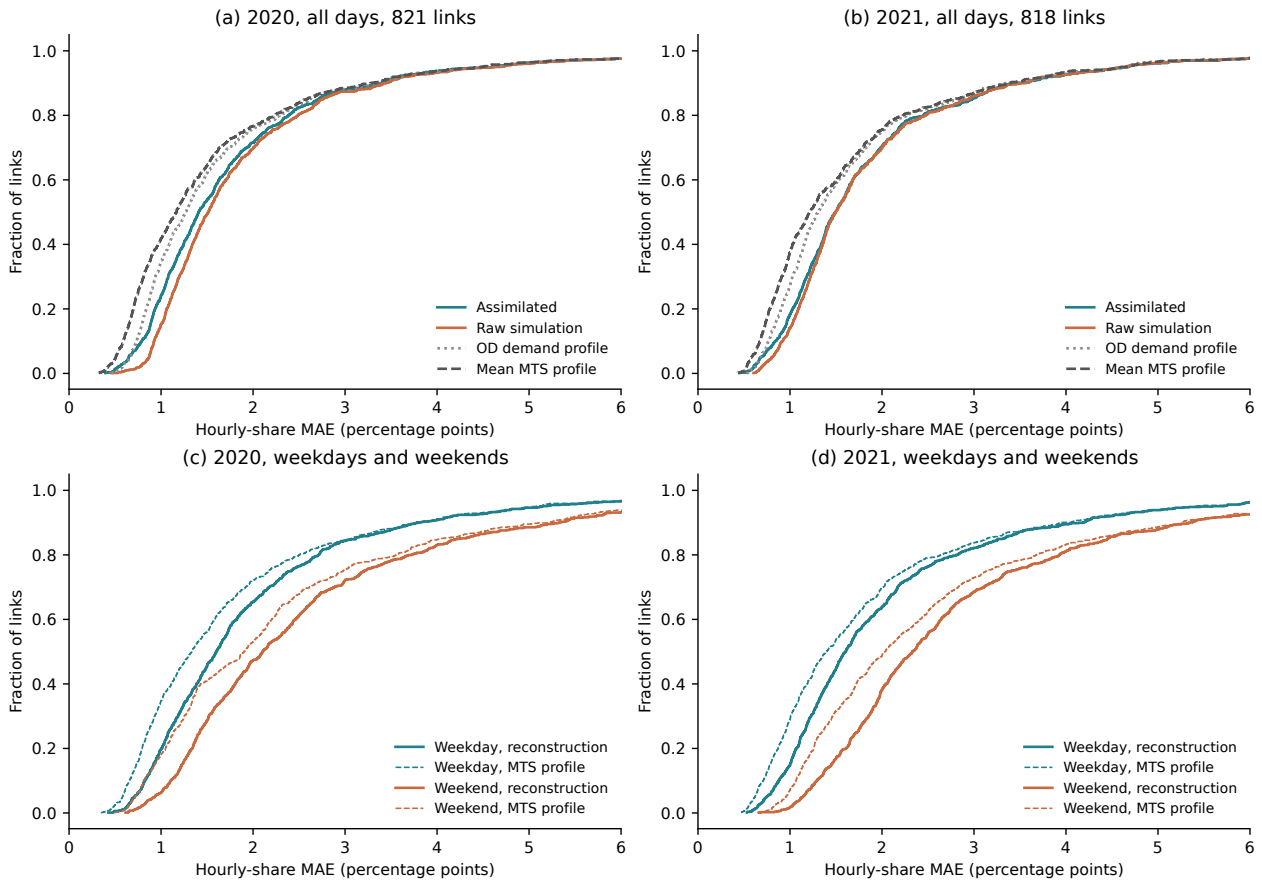


Fig. 5 Hourly-share validation on all usable matched links. (a,b) Cumulative distributions of the link MAE for the raw and assimilated reconstruction, the OD demand profile and the mean MTS profile. **(c,d)** Assimilated reconstruction and MTS profile on weekdays and weekends; curves extend beyond the 6 pp limit.

2020 (136 coils) and 0.36 in 2021 (128 coils), against 0.27 and 0.28 with shortest-distance routing (Fig. 6). The paired bootstrap difference is 0.09 (95% interval 0.01–0.17) in 2020 and 0.08 (−0.01–0.17) in 2021; the 2021 campaign is out of sample for the route-choice weights, which were estimated in February 2020. The median absolute log-ratio between simulated and observed volumes falls from 0.50 to 0.46 and from 0.51 to 0.48. The level of the simulated volumes is not calibrated to counts: at the coils it is about half of the observed one in both campaigns (ratio 0.52).

2.6 Reach of the detector correction

The correction adds to each coil link the difference between observed and simulated counts and propagates it along the simulated turning shares. Its reach was measured on the GPS links and on withheld coils. On GPS links, the median reduction of the hourly-share error is largest on the coil links themselves (0.40 pp in 2020, 0.22 pp in 2021), 0.19 pp one link away in 2020, 0.06–0.10 pp two and three links away, and zero beyond three links, where the correction does not act (Fig. 7d). Timing information therefore propagates over a few links.

Volumes do not. In a leave-one-detector-out test, each coil with complete data is removed from the correction in turn and its observed counts are compared with the reconstruction on its link (152 coils and 614 coil-days in 2020, 140 coils and 571 coil-days in 2021; Fig. 7c). The correction reaches 64% and 57% of the withheld coils; their median error falls from 72% to 68% and from 72% to 69% of the observed volume, with 45% and 39% of the coils improved and 19% worsened, but the pooled absolute error rises from 78% to 88% and from 76% to

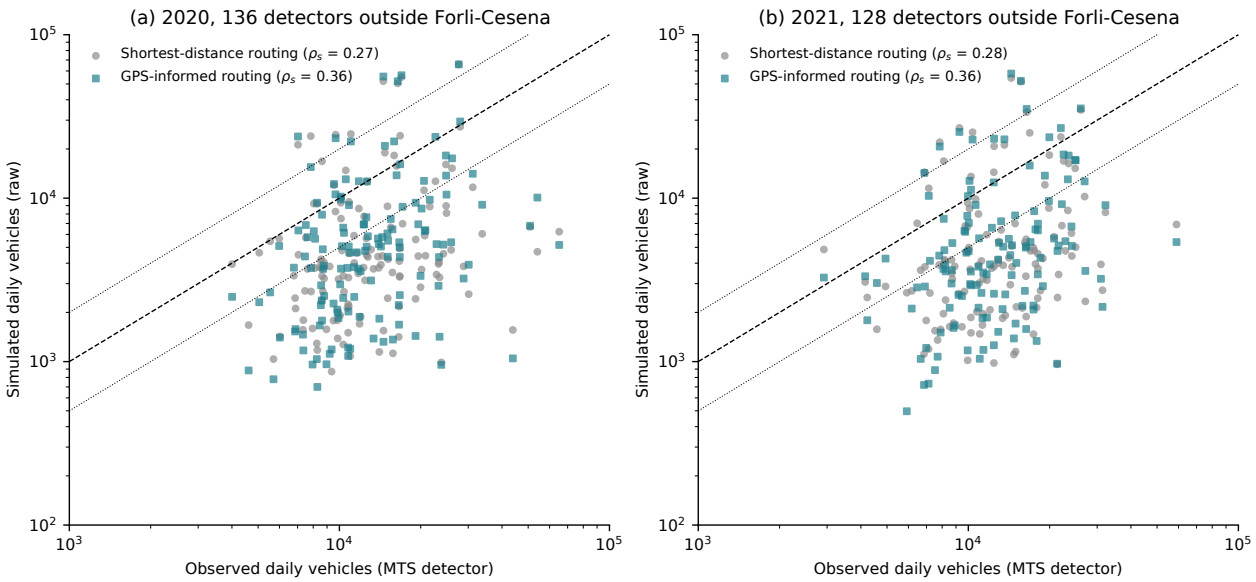


Fig. 6 Daily volumes at detectors outside Forlì–Cesena. Raw simulated against observed daily vehicles (mean over complete days) for shortest-distance routing and GPS-informed routing, **(a)** 10–17 February 2020 and **(b)** 28 June–4 July 2021; ρ_s is the Spearman rank correlation. Dashed line: equality; dotted lines: factor of two.

84%, driven by large increments that reach low-volume links. Limiting the propagation depth reduces this effect: on a fixed set of withheld coils in 2020 the pooled error increases by 5.7 pp with propagation over one link and by 13.6 pp over three links (Supplementary Section S4). The correction is therefore best used locally, at and immediately around instrumented sections.

3 Discussion

The reconstruction provides an hourly value on every directed link of the regional network, and the three tests show which of its properties are supported by observation. The timing of traffic is: simulated and observed peaks coincide and the median link profile differs from the GPS one by about a third of the mean hourly share. It is not better than simpler profiles, however. A mean detector profile describes the hourly shape of traffic on the GPS links more accurately, consistent with the strength of historical profiles in traffic forecasting [44, 45], and the injected demand profile is also more accurate than the simulated one. The value of the simulation lies in the spatial allocation of the demand, which no single profile provides, and its validation requires volume data.

The volume test shows that the spatial allocation depends strongly on route choice. With uniform transition probabilities and shortest-distance paths, agents spread over secondary roads and the volumes of the main roads are underestimated most. Speeds and road-class usage measured by GPS in a single province, transferred to the whole region by road class, improve the rank agreement with detectors in the other provinces and in a later season. This is a modest gain: a rank correlation of 0.36 leaves most of the variation in link volumes unexplained, and the absolute level is not calibrated. Joint estimation of demand and route choice from counts [19, 21, 22] is the natural next step, and the volume test introduced here provides the hold-out criterion for it.

The correction behaves differently for timing and for volume. Increments derived at a detector improve the hourly shape of neighbouring links, but when they are propagated as absolute vehicle numbers they overshoot on links whose volume differs from that of the detector link. The leave-one-out test therefore favours a local

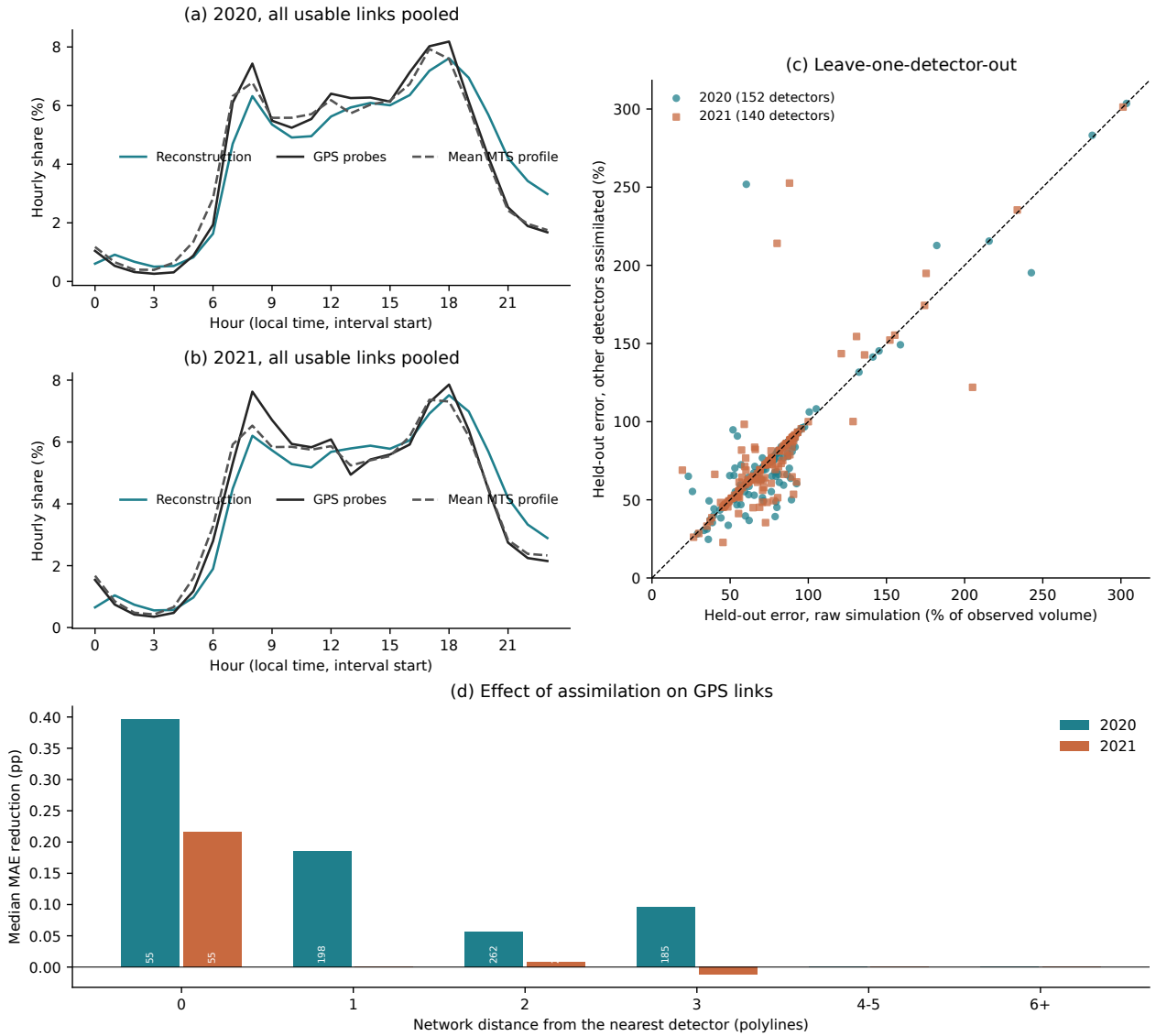


Fig. 7 Timing and reach of the correction. (a,b) Hourly shares pooled over all usable links: reconstruction, GPS probes and mean MTS profile. Pooling weights links by their own volumes, so these curves are not regional traffic shares. **(c)** Error on each withheld coil, as a percentage of its observed volume, with the raw reconstruction and with the other coils assimilated; dashed line, equal errors. **(d)** Median reduction of the hourly-share error on GPS links by network distance from the nearest coil, in polylines; numbers of links inside the bars.

correction, and the reconstruction away from detectors should be read as a model estimate whose spatial ranking, not its absolute level, has been validated.

These conclusions refer to the data that support them. GPS profiles test the hourly shape on links concentrated in one province; volumes are tested at the detector sites, which lie mostly on primary roads; and the demand is the January 2020 OD matrix in every run, including the summer 2021 campaign, when Italian mobility had changed under the pandemic [46, 47]. The similar performance in the two campaigns indicates that the hourly structure of the demand is more stable than its level. A demand basis covering several seasons, a route-choice model estimated jointly with demand, and the uncertainty of link volumes are the main extensions; the evaluation framework applies unchanged to them.

4 Methods

4.1 Data and network

The network is the project graph derived from OpenStreetMap [48, 49], whose road coverage has been estimated to exceed 95% in most European countries [50]. Polylines store connectivity, length, direction and turn restrictions. Each polyline was assigned the OpenStreetMap class of the ways nearest to three points along it (within 30 m; Geofabrik extract of 26 October 2026, ODbL). Municipality OD activity from January 2020 is averaged separately for Monday–Friday and Saturday–Sunday; the public holidays of 1 and 6 January enter the weekday average. Each hourly OD value is injected uniformly in the corresponding hour of local time. MTS counts are available every 15 min in UTC and are converted to local time. For 2020 we use the project archive of the regional counts; for 2021 we use the regional open-data release of the same counts (Regione Emilia-Romagna, CC BY), which in local time reproduces the archive for February 2020.

4.2 Simulation and route choice

The `minimocas` agent-based model advances in 10 s steps and writes hourly outputs. The speed of an agent is the speed of its link, reduced above a critical density. Source–destination agents follow best paths computed with a Dijkstra engine [51] that respects turn restrictions; random-endpoint agents choose the next link at each node with probability proportional to its transition weight. The cost of link i in the best-path computation is

$$w_i = L_i \left[\alpha + (1-\alpha) \frac{1/c_i - 1/c_{\max}}{1/c_{\min} - 1/c_{\max}} \right], \tag{1}$$

where L_i is its length, c_i its usage weight and $\alpha = 0.5$. Link speeds and usage weights are assigned by road class (Table 1); usage weights are proportional to the median daily GPS probes per covered link of the class, and the same weights define the transition probabilities. Assignment is not equilibrated [12], and the model does not resolve the congestion phenomena of microscopic traffic-flow models [52–54]. In two of the fifteen simulated days, 50 next-node decisions in total had no feasible path under turn restrictions with the cost of Eq. (1) and used the shortest-distance path instead. The simulator records, for each hour and direction, the number of distinct agents on each polyline; virtual barriers at the coil locations count point crossings.

GPS speeds were obtained by matching every fix with engine on and speed of at least 5 km/h to the nearest polyline within 20 m whose bearing agrees with the heading within 30°, and taking the median of weekday fixes between 10:00 and 16:00. Route-choice variants (shortest distance, travel time, mixed, and usage-weighted) were compared for 10 February 2020 by the Spearman correlation between simulated daily volumes and GPS counts on the links of Forlì–Cesena.

4.3 Detector correction

Let $u_i(h)$ and $b_i(h)$ be the unique-link and virtual-barrier counts on the link of coil i in hour h, and $c_i(h)$ the observed count after direction mapping. We compute

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \qquad \Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}, \qquad \delta_i(h) = u_i(h)\Delta_i(h), \tag{2}$$

and set $\widetilde{u}_i(h) = u_i(h) + \delta_i(h)$, which equals $c_i(h)$ whenever $u_i, b_i \geq 1$. The increment is propagated downstream and upstream: each adjacent link receives the increment of its parent multiplied by its own simulated turning share, $\delta_{(n_2,n)} = \delta_{(n_1,n_2)}\, u_{(n_2,n)} / \sum_m u_{(n_2,m)}$. Propagation stops when the increment falls below one vehicle, after three links along a path, at another coil link, at a forbidden turn or at a closed link; negative counts are set to zero.

4.4 Validation

For each matched polyline i the ratio between probes and simulated vehicles over the same days is

$$\eta_i = \frac{\sum_{d,h} \psi_i(d,h)}{\sum_{d,h} \phi_i(d,h)}, \tag{3}$$

where ψ_i counts probes and ϕ_i simulated vehicles. We compare hourly shares $p_i(h)$ and $q_i(h)$ of the simulated and GPS counts summed over the campaign days, both directions together, through

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} |p_i(h) - q_i(h)|, \tag{4}$$

in percentage points; links with zero simulated or GPS total are excluded. GPS counts come from the map matching of the original extracts; we use the links whose geometry is identical in the project graph. The OD baseline sums the creation rates of each hour; the MTS baseline averages the normalized hourly profiles of the coil directions complete on all days of the subset; the uniform baseline assigns 1/24 to every hour.

Daily volumes are compared at coils with complete data in both directions, averaging over complete days; the Spearman difference between routing variants is tested with 5,000 paired bootstrap resamples of coils (seed 20261028). In the leave-one-detector-out test the withheld coil is removed from the correction and from the links that stop propagation, and the error is $\sum |\tilde{u}(h) - c(h)| / \sum c(h)$ over hours, directions and days. The depth experiment uses a fixed random 20% of the complete coils of each 2020 day as test set (seed 20261027). Network distance from the nearest coil is the minimum number of polylines between a link and any coil link.

Supplementary Material

The Supplementary Material describes the data layers and the demand classes (S1), the simulation model (S2), the route-choice variants (S3), the detector correction and its sensitivity to propagation depth (S4), additional validation results (S5) and the reproduction of all results (S6).

Availability of data and materials

Mobile-phone and Viasat GPS data are proprietary and cannot be redistributed. MTS loop-detector counts are published by Regione Emilia-Romagna under a CC BY licence (open-data catalogue, dataset "Sistema MTS – Flussi di traffico al quarto d'ora"). Road classes are derived from OpenStreetMap (© OpenStreetMap contributors, ODbL). Aggregate results are included in the code repository. **[TODO: Confirm the route for access to the restricted data for review.]**

Code availability

The simulator, the correction and all analysis scripts are in https://github.com/federicobellisardi/multi-sensor-traffic-reconstruction (directory `article1/analysis`). **[TODO: Make the repository public, confirm the licence of `minimocas` and archive a release with a DOI before submission.]**

Ethics approval and consent to participate

The inherited manuscript describes operator-provided, anonymized and aggregated mobile-phone counts, with no individual-level records accessed by the authors. **[TODO: Authors must verify the delivered aggregation level, applicable permissions and the final ethics/consent declaration.]**

Competing interests

C.M. and A.F. are employed by Mindicity, which contributed to the modelling framework. **[TODO: Confirm this inherited declaration and the absence or presence of other interests for every author.]**

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Authors' contributions

The existing author list is preserved. **[TODO: Confirm contributions to this new manuscript and the additional analyses; no final-author approval is asserted by this working draft.]**

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